



Where We Stand In The Race To Quantum Computing Technology

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Quantum entanglement is a process by which microscopic objects like electrons or atoms lose their individuality to become better coordinated with each other. Entanglement is at the heart of new quantum technologies that can lead to great advances not only in computing but also in communication and sensing.

These entangled states are very fragile; even a tiny disturbance will undo the entanglement. Hence, it is important for current quantum technologies to physically isolate the microscopic systems they work with. These typically operate at temperatures close to absolute zero or -273 degrees Celsius.

What are entangled states? Pairs or groups of particles that are entangled can become so intertwined that measuring the properties (through the state) of one will instantly change the properties of others. Quantum computers must have several dozen of these entangled states or qubits to become a commercially viable computing method.

A quantum computer is a machine that performs calculations based on the laws of quantum mechanics, which is the behaviour of particles at the sub-atomic level. The excitement surrounding quantum computing is mainly due to the fact that it reduces the computational complexities of certain problems. Only very special classes of algorithms are known to enjoy quantum speedup. Based on the evaluation of publicly available information regarding progress to date in the field of quantum computing, there appears no fundamental reason why a large, fault-tolerant quantum computer could not be built in principle.

There's a race to build a machine that can outperform a classical one. In the current industrial or commercial domain, we have no quantum machine that can outperform a classical one. The goal, however, is to build a quantum machine that can solve problems faster than classical computers. This is called quantum supremacy or quantum ascendancy.

The below-mentioned technologies are considered appropriate and relevant

